

Customer Churn Analysis –Summary & Recommendation

Objective

The primary objective of this project was to analyze customer churn behavior and identify the major factors contributing to customer attrition. By performing exploratory data analysis (EDA) and creating multiple visualizations, the project aimed to uncover patterns related to customer demographics, services, contract types, payment methods, and tenure that influence churn decisions.

Overall Churn Analysis

The analysis revealed that approximately:

- **73.46%** of customers were retained
- **26.54%** of customers churned

This indicates that nearly **1 out of every 4 customers** discontinued the service, making churn prediction and retention strategies highly important for the business.

Key Insights

1. Contract Type and Churn

One of the strongest churn indicators was the customer's contract type.

Findings:

- Customers with **Month-to-Month contracts** showed the **highest churn percentage**
- Customers with **One-Year and Two-Year contracts** were significantly less likely to churn

Business Insight:

Long-term contracts increase customer retention and reduce churn probability.

Churn Rate by Tenure

Customer tenure played a major role in churn behavior.

Findings:

- Customers with **low tenure (new customers)** had the highest churn rates
- Customers staying for a longer duration were more stable and loyal
- Churn gradually decreased as tenure increased

Business Insight:

The company should focus on improving onboarding experience and customer engagement during the initial months of subscription.

Internet Service Analysis

The analysis of internet service types showed major differences in churn behavior.

Findings:

- Customers using **Fiber Optic internet** had noticeably higher churn rates
- DSL users showed comparatively lower churn percentages
- Customers without internet service had the lowest churn behavior

Business Insight:

Fiber Optic customers may face issues related to pricing, expectations, or service quality, requiring targeted service improvements.

Service-Based Churn Analysis

Several additional services strongly influenced customer retention.

Customers without the following services had higher churn rates:

- OnlineSecurity
- TechSupport
- OnlineBackup
- DeviceProtection

Business Insight:

Customers using bundled and support-oriented services are more likely to remain with the company. Promoting these services could improve retention.

Payment Method Analysis

Payment behavior also affected churn trends.

Findings:

- Customers using **Electronic Check** payment methods showed higher churn percentages
- Customers using automatic payment methods had better retention

Business Insight:

Improving digital payment experience and encouraging automatic payment options may help reduce churn.

Demographic Analysis

Senior Citizen Analysis

Findings:

- Senior citizens showed a comparatively higher churn percentage than non-senior customers

Business Insight:

The company may need customer support strategies specifically tailored for senior customers.

Visualization Summary

Multiple visualizations were created throughout the analysis to better understand customer behavior and churn patterns.

Visualizations Used:

- Count Plots
- Stacked Bar Charts
- Pie Charts
- Histograms
- Correlation Analysis

Purpose of Visualizations:

These charts helped identify:

- Service usage patterns
- Customer retention trends
- Churn distribution across categories
- Relationships between customer features and churn

The visualizations clearly demonstrated that customers with fewer services, shorter tenure, and flexible contracts were more likely to churn.

Overall Conclusion

The customer churn analysis identified several major drivers of churn, including:

- Short tenure
- Month-to-Month contracts
- Lack of support/security services
- Fiber Optic internet usage
- Electronic check payments

The findings suggest that the company can reduce churn by:

- Encouraging long-term contracts
- Improving customer onboarding
- Enhancing Fiber Optic service quality
- Promoting bundled support services
- Creating targeted retention campaigns for high-risk customers

Overall, this analysis provides valuable business insights that can support data-driven decision-making and improve customer retention strategies.